

Estimation of direct normal irradiance from measured global and corrected diffuse horizontal irradiance



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ABSTRACT

DNI (direct normal irradiance) can be calculated from global and diffuse horizontal irradiance measurements. However, the diffuse irradiance values need to be corrected because the pyranometer's shadowband does not obstruct only the solar disk but also a larger part of the sky vault. In this study, we use four diffuse correction models (Drummond [1], LeBaron et al. [2], Batlles et al. [3], Muneer and Zhang [4]) and, considering the importance of DNI for several solar energy projects, we try to assess the performance of the models when the calculated DNI is compared to the measured one by the pyrhe-liometer. Based on 1-min averaged measurements covering a one-year period in Athens, Greece, it is concluded that the empirical approach of Batlles et al. performs best. It presents the lowest residuals ($\pm 1\%$) with no dependence from the clearness index. The estimated annual DNI, derived from the synergetic use of this model with measurements of horizontal global and diffuse irradiance, is lower by 0.4%, while the calculated values from the global and uncorrected diffuse irradiances are overestimated by 7.7%.

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1. Introduction

Solar irradiance reaching the earth's surface has two components; the direct coming from the sun disc and the diffuse, as a result of scattering of the direct component from the atmospheric constituents. GHI (global horizontal irradiance), the vector sum of direct and diffuse components, is measured nowadays in many stations around the globe. Not all of these stations measure diffuse irradiance while DNI (direct normal irradiance) is measured in a limited number of stations. However, the accurate assessment of DNI reaching the earth's surface is necessary for several applications related to the design, simulation and performance assessment of several solar energy systems and plants but also passive solar buildings. DNI is very sensitive to the aerosol optical depth and is strongly affected by clouds covering the Sun. As a result, DNI exhibits higher spatial and temporal variability than GHI and complicates the accurate simulation of solar power plants or other systems requiring DNI as input.

A common way to calculate DNI derives from synchronized GHI and DHI (diffuse horizontal irradiance) measurements and taking

into account the incidence angle of the sun rays using the following equation:

$$\text{DNI} = \frac{\text{GHI} - \text{DHI}}{\cos(\theta_z)} \quad (1)$$

where θ_z is the solar zenith angle.

GHI is measured using a pyranometer on a horizontal surface. DHI is also measured using a pyranometer positioned horizontally on a support equipped with an adjustable device that blocks the direct component from the sensor. The most commonly used shading devices are shadowbands and shade disks, painted black in order to reduce any reflection towards the sensor. Tracking shadow disks are sometimes used to block the direct component of the sun. These devices may require a little correction [5] and can provide accurate diffuse radiation measurements; however their operation requires a relatively expensive tracking device and presents drawbacks similar to those associated with the pyrhe-liometer.

A fixed shadowband is the most widely used technique in measuring DHI. It requires manual adjustment for Sun declination in the north–south axis every few days and shadows the sensor constantly so that diffuse measurements are being recorded for extended periods of time. However, an inherent problem to the use of the shadowband is that at any time it screens not only the DNI but also a small portion of the DHI. Because of the anisotropy of

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scattered irradiance, with its maximum closer to the sun, this shaded part corresponds to an essential amount of the DHI under cloud-free or partly cloudy skies. Therefore, a correction factor must be introduced to the DHI measurements to compensate for the part of the sky vault obstructed by the shadowband. In more recent studies perforated, non-rotating shadowband is used to intermittently measure both global and diffuse solar irradiance [6].

Several correction models have been proposed [1–5,7–9] to compensate for the DHI component obstructed by the shadowband. These models were usually applied and evaluated with measurements of the site in which the model was developed. Generally these models were taking into account the geometry of the shadowband and were based on assumptions regarding the sky radiance distribution. The latter is affected significantly from the solar zenith angle, the aerosol optical properties and cloudiness [9–14]. Use of fixed shadowbands to block the sun may result either in additional errors or in compensation errors, depending on instrument and shadowband correction algorithm [15].

Historically, Drummond [1] developed a theoretical model based on geometric calculations for the fraction of the sky vault obstructed by the shadowband. The proposed formula could be applied anywhere in the world in order to estimate the amount of DHI blocked by a polar geometry shadowband. In order to derive his correction factor, Drummond assumed an isotropic sky radiance distribution, but after applying this geometric correction to a large amount of data in South Africa, he observed that a further correction of 7% and 3% was necessary for cloud-free and overcast skies respectively in order to take into account the anisotropic radiance distribution.

Several investigators stressed the importance of applying a correction factor on diffuse irradiance measurements due to the anisotropic distribution of sky radiance. Painter [7] performed synchronized measurements of DHI by using a ring and a motor-driven occulting disc. Based on the differences between the two diffuse measurements, a correction factor, necessary to equate the ring and disk measurements under various sky conditions, was proposed. The results were found to be in good agreement with the results of Drummond. Kasten [16] developed an empirical model that used cloudiness, turbidity and declination parameters of the Sun to characterize the sky conditions. Dehne [17] observed that the attempt to transfer these functions to other sites was unsatisfactory. Steven [18], based on measurements of the diffuse-to-global irradiance ratio for characterizing the sky radiance anisotropy, developed a simple two-parameter model to describe the relationship between the isotropic background and the circumsolar component for several types of sky conditions. Stanhill [8] and Kudish and Ianetz [19], applied Steven's model on data from two sites in Israel and proposed that the anisotropic sky corrections, calculated at different times of the year, were up to 30% above the Drummond's geometric corrections.

LeBaron et al. [2] suggested a model that used four parameters to describe both isotropic and anisotropic sky conditions. Each parameter was classified into four categories, thus creating a model that classified the different sky conditions into 256 unique categories. Batlles et al. [3] used the same parameters as LeBaron et al. [2] in order to perform a multiple linear regression analysis. Their model consisted of four equations functions of three of the LeBaron et al. parameters. The choice of the appropriate equation for the calculation of the correction factor was determined by the range of the values of the fourth parameter. More recently, Muneer and Zhang [4] developed and validated a new model based on the work of Moon and Spencer [20] for sky radiance distribution.

Comparative studies were made [21–23,25] in order to assess the performance of the diffuse correction models considering both isotropic and anisotropic conditions as proposed by LeBaron et al., Batlles et al. and Muneer and Zhang with that of Drummond, which

considers only the geometric aspect and is valid only for isotropic conditions. López et al. [21] evaluated the performance of the four correction methods using hourly data from two sites in Bracknell, UK and Beer Sheva, Israel. They found that the use of an isotropic correction factor leads to an overestimation of DHI by 4–11% and they concluded that the LeBaron and Muneer–Zhang models were performing better than the Drummond's and Batlles. They also made a comparative study with the same models for Almería, Spain [22] and concluded that the use of an isotropic correction factor leads to an underestimation of the actual diffuse of 10.2%. The use of anisotropic correction schemes reduced the above underestimation to 8.7%, 2.1% and 4.9% for the LeBaron, Batlles and Muneer's models respectively. Kudish and Evseev [23] attempted to validate the performance of the four models under different sky conditions, based on a one-year hourly data from Beer Sheva, Israel; but did not arrive to a definite conclusion regarding the overall relative predictive ability of the models. However, they stated that, taking into account an overall average performance, the model of Muneer–Zhang performs better than the others. They also utilized the corrected horizontal diffuse irradiance to assess 11 models that predict the diffuse irradiance on a south-facing surface tilted at 40° [24]. Sánchez et al. [25] used one-year hourly averages of DHI, GHI and DNI from Babajoz, South-West Spain and evaluated six correction models. After adjusting the models to the local conditions, their analysis revealed that the empirical approaches developed by LeBaron et al. and Batlles et al. performed better and showed no dependence from the solar zenith angle and the clearness index. In an ulterior study they used these models to correct ultraviolet diffuse irradiance [26]. Results revealed that some aspects of the correction proposed for total diffuse radiation are not suitable for ultraviolet diffuse radiation.

The purpose of this study is to provide an estimation method of the DNI using GHI and corrected DHI data. So, going a step forward from previous works, the results do not focus only on proposing a best model for the correction of DHI due to the shadowband effect but focus also on the uncertainties of the estimated DNI from collocated GHI and DHI measurements. In contrast with previous studies and due to the high temporal variability of DNI, we used 1-min averages of synchronized measurements of DHI, DNI and GHI. We applied four models (Drummond, LeBaron, Batlles and Muneer–Zhang) and we studied their relative performance not only regarding the DHI component but also regarding the calculated DNI under different sky conditions. Preliminary results have been published in international conference [27].

2. Data and methodology

The measured data used in this study were recorded by the radiometric station of the National Observatory of Athens, Greece, during the year 2001. The station is located at the northeast of Athens on a hill of the Penteli mountain (38°N, 24°E, 500 m a.s.l.). The dataset consists of 1-min averaged measurements of GHI, DHI and DNI. The GHI and DHI were measured using a CM11 pyranometer; the pyranometer used for the DHI measurements was shaded using a CM121 shadowband having a 620 mm diameter and a 55 mm width. The DNI was measured using a CH1 pyrliometer, mounted on a Kipp&Zonen Solys 2 solar tracker. All instruments were new and installed for the first time in 2001 therefore their original calibration factors (provided by the manufacturer Kipp&Zonen [28]) were used.

Before calculating the DNI using Eq. (1), a quality control has been applied on all data in order to detect and remove any erroneous values or outliers. Measurements corresponding to $\theta_z \geq 70^\circ$ were removed considering that the angular inaccuracy of the pyranometer exceeds 2% [28]. As defined by López et al. [21], a set of

quality control filters was also applied. Table 1 lists the filters applied, with G_g the GHI, $G_{sc} = 1366.1 \text{ W/m}^2$ the solar constant [29], G_b the DNI, G_d the measured uncorrected DHI, G_{dt} the “true” DHI estimated from synchronized GHI and DNI measurements as defined by López et al. [21,22] and Kudish and Evseev [23], k the diffuse fraction G_d/G_g on a horizontal surface and k_t is the clearness index, defined as:

$$k_t = G_g / [G_{sc} \cos(\theta_z)] \quad (2)$$

The estimated DNI (G_{bi}) derived from GHI and corrected DHI measurements using the equation:

$$G_{bi} = \frac{G_g - G_{di}}{\cos(\theta_z)} \quad (3)$$

where i denotes the measured uncorrected (u), or the estimated DHI values by the Drummond (D), LeBaron (L), Batlles (B) and Muneer–Zhang (M) shadowband correction models.

Table 1 provides also the remaining number of data values after the application of each filter. The original dataset consisted of 525,600 1-min measurements, 262,812 of which were the daytime records ($\theta_z \leq 90^\circ$); after applying the filters, 120,948 measurements remained for further processing i.e., 46% of the original diurnal database. As expected, the stricter filter was that of $\theta_z \leq 70^\circ$, which removed almost 31% of the initial database. The second stricter filter was in “true” diffuse irradiance (G_{dt}) that takes into account errors introduced during the calculation of the “true” diffuse irradiance from global and DNI irradiance. Negative values that occurred after calculating the DNI from GHI and DHI measurements were attributed, based on the meteorological observations at the site, to rainy weather conditions when the sky was completely overcast by thick clouds. Under those conditions, GHI and DHI values were very close and negative DNI values were derived due to the measurement uncertainties of the pyranometers. This result was also confirmed by the zero DNI measurements at the same time.

The estimated DNI values were analyzed under different cloud conditions and assessed both graphically and statistically. The evaluation was based on regression analysis and the statistical indices used to numerically quantify the performance of the models were the slope of the best fit line, the resulting coefficient of determination (R^2), the RMSE (root mean square error), the rRMSE (relative root mean square error), the MBE (mean bias error), the rMBE (relative mean bias error), and the t -statistic index, which is computed using both the RMSE and the MBE. The statistics are defined by the following expressions [30]:

Table 2

Statistical results for the 1-min values of the direct normal irradiance calculations from corrected diffuse irradiance data without and with the four shadowband correction models for all, clear, partially cloudy and cloudy sky conditions.

Correction models	Slope	R^2	RMSE (W/m^2)	MBE (W/m^2)	t -Statistic
All sky conditions: $N = 120,816$ min, $\overline{G_{b \text{ Meas}}} = 651.73 \text{ W/m}^2$					
Uncorrected	1.057	0.992	60.8 (9.3%) ^a	50.1 (7.7%) ^a	505.08
Drummond	1.029	0.994	39.3 (6.0%)	26.3 (4.0%)	313.32
LeBaron et al.	0.993	0.994	29.7 (4.6%)	−5.0 (−0.8%)	59.62
Batlles et al.	0.996	0.994	29.0 (4.5%)	−2.7 (−0.4%)	32.21
Muneer and Zhang	1.005	0.995	28.6 (4.4%)	8.3 (1.3%)	105.89
Clear sky conditions: $N = 97,446$ min, $\overline{G_{b \text{ Meas}}} = 760.57 \text{ W/m}^2$					
Uncorrected	1.052	0.982	55.6 (7.3%) ^a	45.5 (6.0%) ^a	443.66
Drummond	1.026	0.983	36.2 (4.8%)	23.5 (3.1%)	266.16
LeBaron et al.	0.994	0.984	27.9 (3.6%)	−5.8 (−0.7%)	67.10
Batlles et al.	0.996	0.981	28.1 (3.7%)	−3.1 (−0.4%)	34.71
Muneer and Zhang	1.003	0.985	25.3 (3.3%)	4.4 (0.6%)	54.97
Partially cloudy sky conditions: $N = 18,158$ min, $\overline{G_{b \text{ Meas}}} = 245.92 \text{ W/m}^2$					
Uncorrected	1.200	0.979	86.7 (35.2%) ^a	79.5 (32.3%) ^a	311.24
Drummond	1.110	0.980	55.2 (22.4%)	44.3 (18.0%)	181.87
LeBaron et al.	0.981	0.976	38.5 (15.7%)	−8.4 (−3.4%)	30.17
Batlles et al.	0.999	0.979	35.4 (14.4%)	−1.6 (−0.6%)	6.17
Muneer and Zhang	1.062	0.981	42.7 (17.3%)	27.6 (11.2%)	114.47
Cloudy sky conditions: $N = 5212$ min, $\overline{G_{b \text{ Meas}}} = 30.74 \text{ W/m}^2$					
Uncorrected	1.629	0.887	43.8 (142.5%) ^a	35.1 (114.1%) ^a	96.51
Drummond	1.336	0.907	27.2 (88.5%)	16.8 (54.5%)	56.51
LeBaron et al.	1.286	0.896	28.5 (92.0%)	22.3 (72.5%)	90.11
Batlles et al.	1.089	0.917	18.3 (59.6%)	1.7 (5.6%)	6.88
Muneer and Zhang	1.293	0.907	25.4 (82.7%)	15.0 (48.8%)	52.76

^a RMSE and MBE as percentage of $\overline{G_{b \text{ Meas}}}$.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i^{\text{corrected}} - x_i^{\text{meas}})^2} \quad (4)$$

$$\text{MBE} = \frac{1}{N} \sum_{i=1}^N (x_i^{\text{corrected}} - x_i^{\text{meas}}) \quad (5)$$

$$\text{rRMSE} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i^{\text{corrected}} - x_i^{\text{meas}})^2}}{\frac{1}{N} \sum_{i=1}^N x_i^{\text{meas}}} \quad (4a)$$

$$\text{rMBE} = \frac{\frac{1}{N} \sum_{i=1}^N (x_i^{\text{corrected}} - x_i^{\text{meas}})}{\frac{1}{N} \sum_{i=1}^N x_i^{\text{meas}}} \quad (5a)$$

where $(1/N) \sum_{i=1}^N x_i^{\text{meas}}$ is the mean value of the measured DNI

Table 1

Quality control filters of solar irradiance.

Irradiance	Filter	1-min values removed	Remaining 1-min values
All ($\theta_z \leq 90^\circ$)	$\theta_z \leq 70^\circ$	82,362 (31.3%)	180,450
Global	$0 < G_g < 1.2G_{sc}$	0 (0%)	180,450
	$0 < k_t < 1$	417 (0.16%)	180,033
Measured diffuse	$0 < G_d < 0.8G_{sc}$	0 (0%)	180,033
	$G_d < G_g$	38 (0.01%)	179,995
	$0 < k < 1$	0 (0%)	179,995
Direct normal	$0 < G_b < G_{sc}$	7 (0%)	179,988
“True” diffuse	$0 < G_{dt} < 0.8G_{sc}$	11 (0%)	179,977
	$0.6G_{dt} < G_d < G_{dt}$	26,660 (10.14%)	153,317
Negative values of corrected DNI		29,402 (19.1%)	123,915
Operational errors			
Disconnected instruments		3099 (2.50%)	120,816

$$t = \sqrt{\frac{(N-1)MBE^2}{RMSE^2 - MBE^2}} \quad (6)$$

The results are summarized in Table 2.

The sky conditions were defined according to clearness index k_t [23]:

$$\text{Cloudy } 0 \leq k_t < 0.35 \quad (7a)$$

$$\text{Partially cloudy } 0.35 \leq k_t \leq 0.65 \quad (7b)$$

$$\text{Clear sky } k_t > 0.65 \quad (7c)$$

In existing literature analysis of solar radiation data is mainly based on hourly irradiance data therefore the analysis was also performed on hourly-averaged values of the same dataset.

3. Description of the shadowband correction models

All models provide a correction factor, C_i , which is applied along with the measured DHI (G_d) data in order to calculate the corrected DHI (G_{di}) using the formula:

$$G_{di} = C_i \times G_d \quad (8)$$

In this study, we present the following four models named after the name of the first author of the paper who introduced them:

3.1. The Drummond model

Drummond was one of the pioneers in proposing shadowband correction methods. He used global and diffuse irradiance data from nine stations located at Pretoria, South Africa, an area much of which lies in the relatively cloud-free anticyclonic belt of the Southern Hemisphere. Drummond's model [1], is thus based on an assumption that the diffuse radiation is isotropically distributed over the sky hemisphere. Due to this assumed diffuse radiation's isotropy, the correction factor can be estimated from the shadowband geometrical features of pyranometer.

The factor correcting the irradiance intercepted by the shadowband was calculated by comparing the sky irradiance measured using the shadowband against the sky irradiance measured by replacing the shadowband with a 10 cm diameter disk placed 1 m above the sensor. The comparisons were limited to cloudless skies. The fraction of the diffuse horizontal irradiance obstructed by the shadowband (X) is given by [21,23]:

$$X = 2b \cos^3(\delta) t_0 \sin(\varphi) \sin(\delta) + \cos(\varphi) \cos(\delta) \sin(t_0) / (\pi r) \quad (9)$$

with b the width and r the radius of the shadowband and

$$t_0 = \cos^{-1}[\tan(\varphi) \tan(\delta)] \quad (10)$$

the hour angle at the sunset ($\theta_z = 90^\circ$) in radians, φ the latitude of the location and δ the solar declination.

The Drummond isotropic correction factor is then expressed as

$$C_D = 1/(1 - X) \quad (11)$$

Drummond also compared the diffuse irradiance measured using a shadowband against the total irradiance for completely overcast skies and concluded [31] that a further correction of 7%, 5% and 3% for cloud-free, partly cloudy and overcast skies respectively was necessary in order to take into account the anisotropic distribution of solar irradiance.

3.2. The LeBaron model

The distribution of scattered solar irradiance changes markedly with sun's elevation, atmospheric turbidity and cloudiness. Isotropic conditions are a satisfactory approach only for overcast skies. Based on two years of hourly data of global, diffuse horizontal and direct normal irradiance from Albany, New York, and Bluefield, West Virginia in order to include some variation in climatic conditions and latitude separation, LeBaron et al. [2] examined the anisotropic contribution to the total shadowband correction for seven different levels of sky cloudiness. They proposed a correction model that uses four parameters describing both isotropic (geometric) and anisotropic (sky conditions) effects. The first parameter, the geometric screening, is the Drummond's correction factor, C_D , while the other three parameters, solar zenith angle θ_z , epsilon ε and delta Δ describe the anisotropic contribution to the final shadowband correction. Epsilon, the clearness index is defined as:

$$\varepsilon = (G_{du} + G_{bc})/G_{du} \quad (12)$$

with G_{du} the uncorrected measured DHI and G_{bc} the DNI calculated from global and diffuse radiation measurements on a horizontal surface.

Delta is the brightness index and is a function of the cloud thickness or aerosol loading. It is defined as:

$$\Delta = G_{du}/(G_{on} \cos(\theta_z)) \quad (13)$$

where G_{on} is the extraterrestrial solar irradiance measured on a plane normal to the radiation and is calculated from the equation:

$$G_{on} = G_{sc} \left(1 + 0.033 \cos\left(\frac{360n}{365}\right) \right) \quad (14)$$

where n is the day of the year (1–365).

LeBaron's model was developed by classifying each of the parameters, θ_z , C_D , ε , Δ , into four categories (i, j, k, l) thus creating a model of 256 categories in total; the LeBaron correction factor, C_L , is looked up from a table [2].

3.3. The Batlles model

Batlles et al. [3] developed two different correction models based on the same parameters as the LeBaron model. Using two-thirds of hourly data from Madrid, Spain, covering a two-year period and the other one-third plus 5-min data from Almería, Spain for validation purposes, resulted to two parameterizations, a simpler one, model A, and a more complex, model B. It appeared that model B was more universal than model A.

The Batlles model B is based on the same parameters as the LeBaron model. However, the proposed correction factor (C_B) is parameterized against ε and is expressed as an analytical function of C_D , θ_z and Δ . The model is described by the following set of equations:

$$\begin{aligned} \varepsilon \leq 3.5 &\rightarrow C_B = 1.178C_D + 0.0207 \log(\Delta) + 0.422e^{-\frac{1}{\cos(\theta_z)}} \\ 3.5 \leq \varepsilon \leq 8 &\rightarrow C_B = 1.454C_D + 0.655 \log(\Delta) + 0.4756e^{-\frac{1}{\cos(\theta_z)}} \\ 8 \leq \varepsilon \leq 11 &\rightarrow C_B = 1.486C_D + 0.495 \log(\Delta) \\ \varepsilon > 11 &\rightarrow C_B = 1.384C_D + 0.363 \log(\Delta) \end{aligned} \quad (15)$$

C_B depends on the solar zenith angle (θ_z) in the two first equations corresponding to scattered and overcast sky conditions. When there are few or no clouds at all (that is for higher ε values), C_B is practically independent from θ_z .

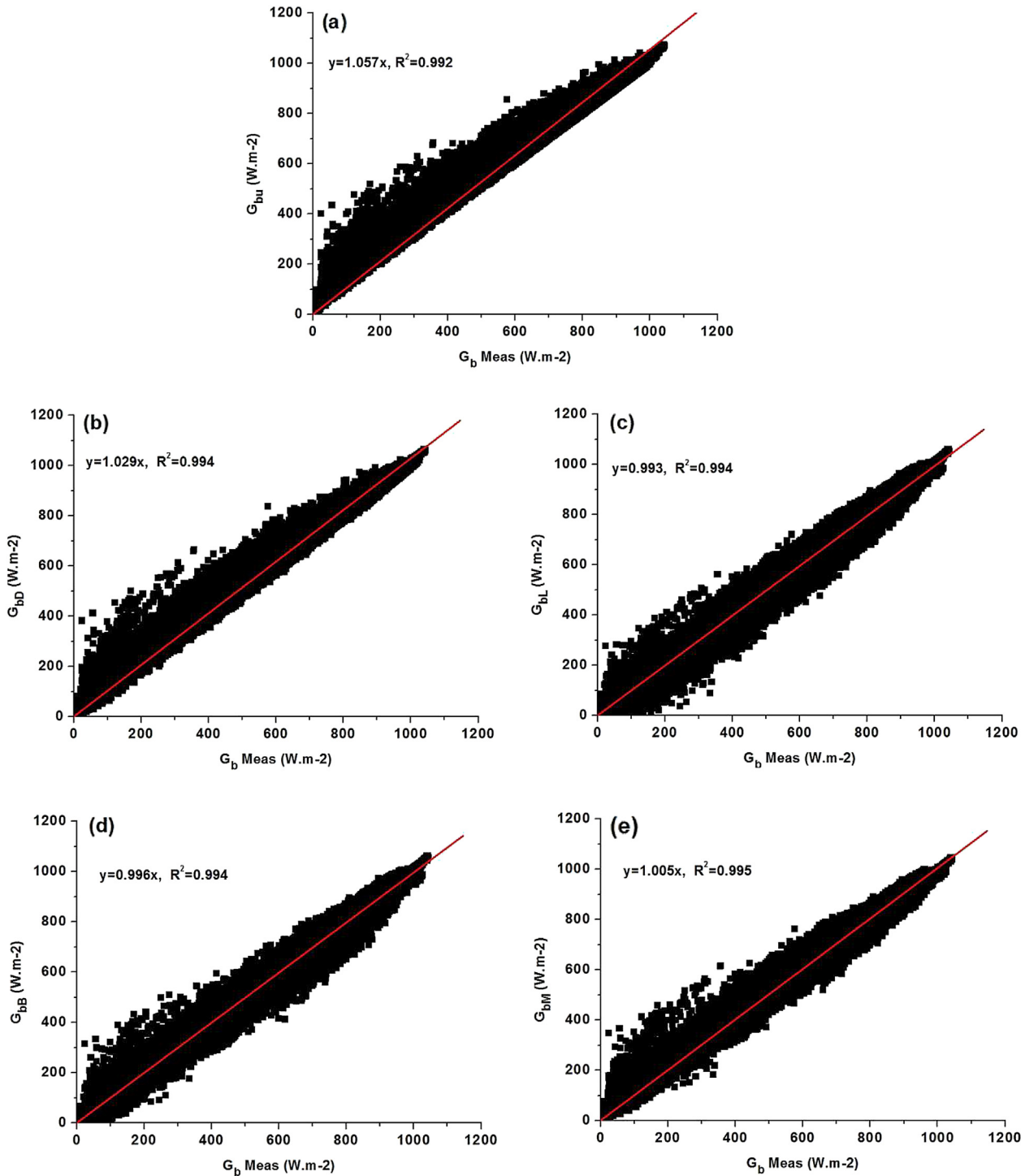


Fig. 1. Calculated values of direct normal irradiance, for all sky conditions, using uncorrected diffuse irradiance measurements (a), Drummond (b), LeBaron (c), Batlles (e) and Muneer–Zhang (d) shadowband correction models versus measurements.

3.4. Muneer–Zhang model

This anisotropic model is based on the use of a single diffuse irradiance distribution index, b , introduced after the work of Moon and Spencer [20]. As a follow up of Muneer's previous work [32], the proposed model [4] was validated using two datasets from sites with completely different sky conditions, Bracknell, UK and Beer Sheva, Israel and compared to the Drummond's model. The data consisted of one year hourly values

of global horizontal, direct normal and diffuse irradiance measured using a shadowband. The clearness index, k_t , was used as a parameter to determine the value of b ; b was then used to obtain a correction factor to account for the diffuse irradiance obscured by the shadowband. Muneer [4,21] derived a relationship between b and k_t presented in Eqs. (16) and (17). Parameters b_1 and b_2 represent the irradiance distribution indices for the two sky quadrants, the one containing the sun disk and the opposite to it

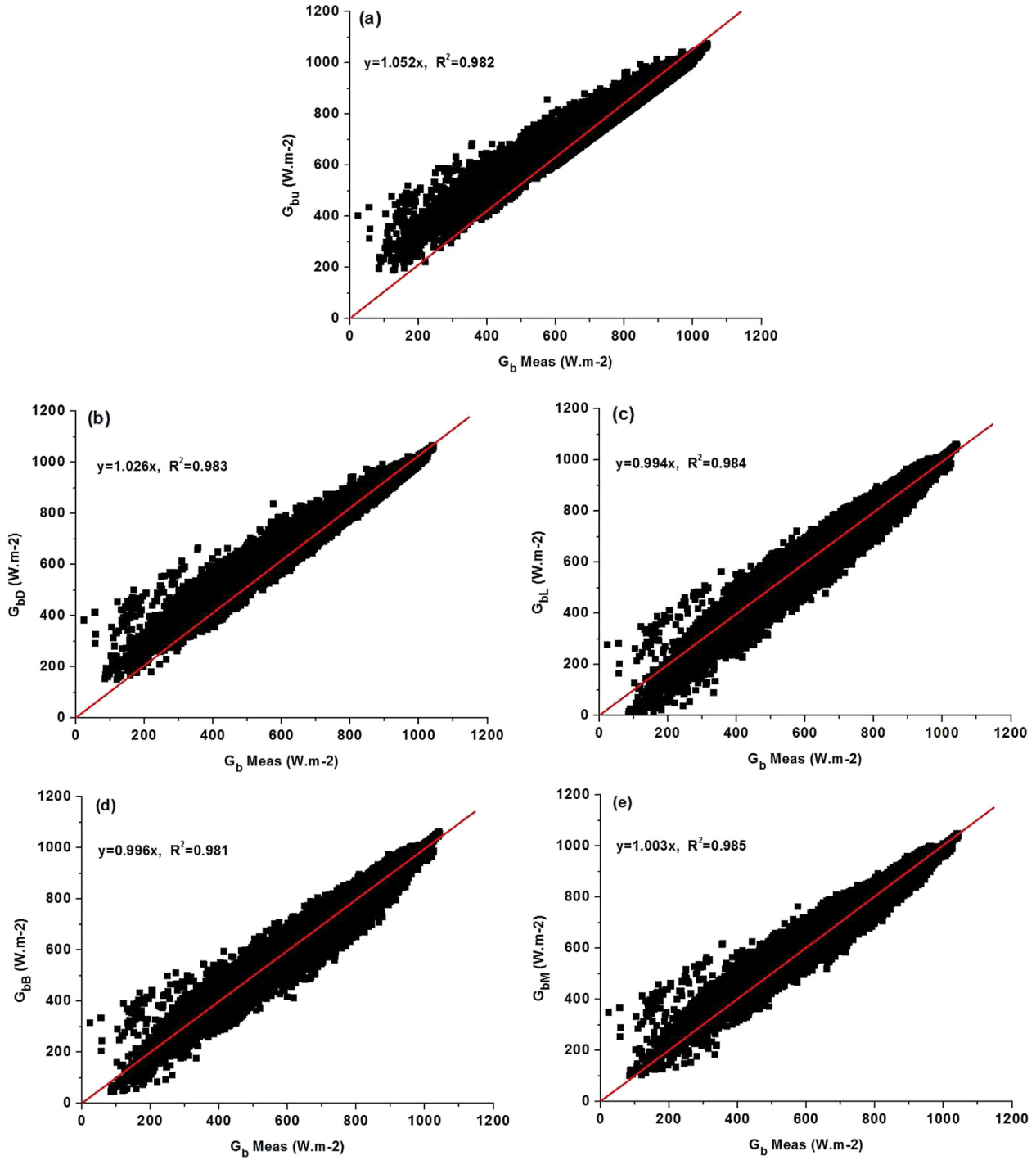


Fig. 2. Same as Fig. 1 but for clear sky conditions.

$$k_t > 0.2 \rightarrow b_1 = \frac{3.6 - 10.462k_t}{-0.4 + 6.974k_t}, b_2 = \frac{1.565 + 0.990k_t}{0.957 - 0.660k_t} \quad (16)$$

and

$$k_t \leq 0.2 \rightarrow b_1 = b_2 = 1.68 \quad (17)$$

with k_t , the clearness index, as defined in Eq. (2).

The correction factor, C_M , is given by the expression:

$$C_M = \frac{1}{1 - \frac{F}{I_d}} \quad (18)$$

The parameter I_d is defined as:

$$I_d = \left(\frac{\pi L_z}{6} \right) \left(\frac{3 + 2b_1}{1 + b_1} + \frac{3 + 2b_2}{1 + b_2} \right) \quad (19)$$

where L_z ($\text{W m}^{-2} \text{sr}^{-1}$) is the zenith radiance.

The parameter F is defined as:

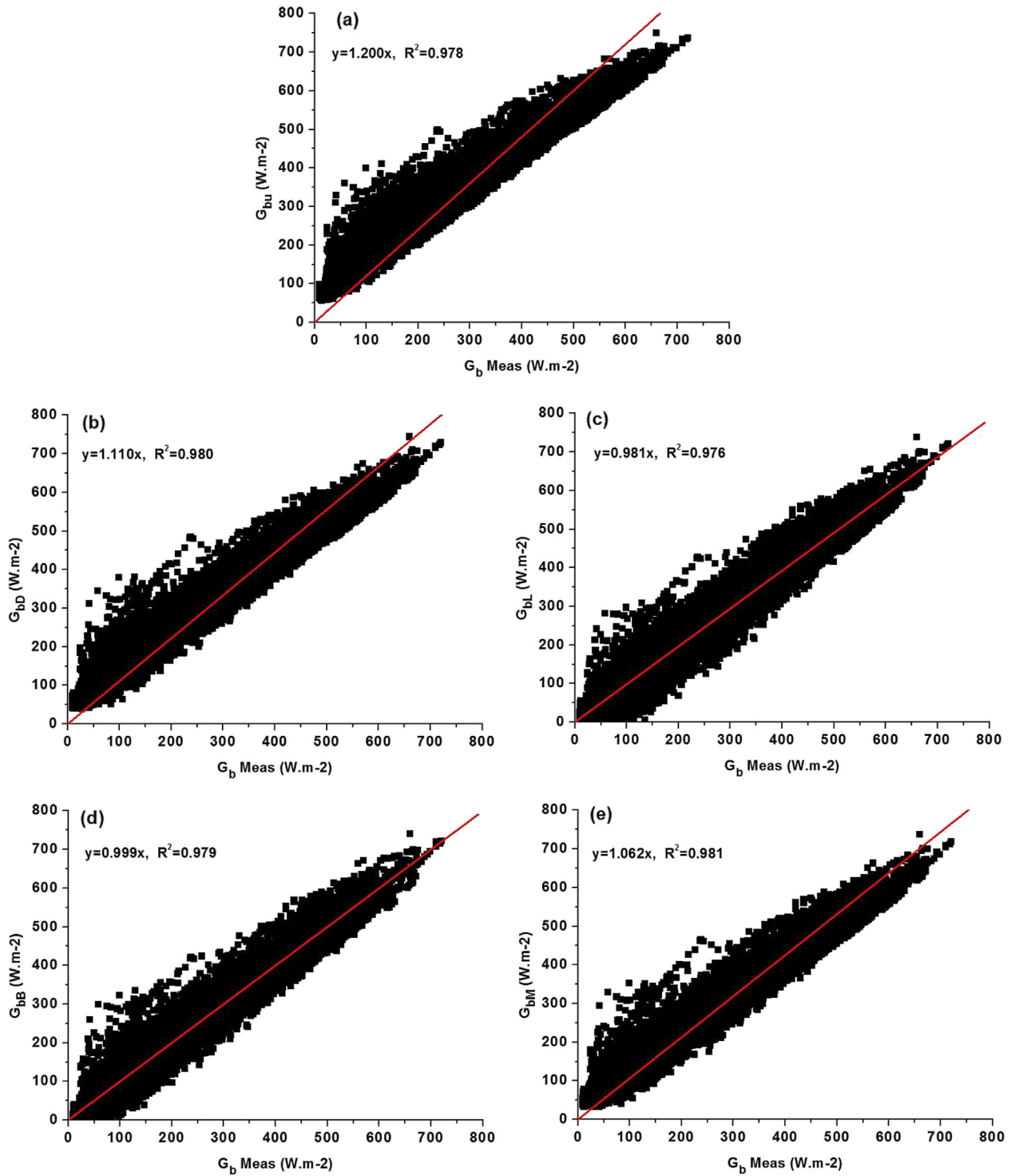


Fig. 3. Same as Fig. 1 but for partly cloudy sky conditions.

$$F = 2wL_z \cos^3(\delta) \left[\frac{I_1 + b_1 I_2}{1 + b_1} \right] \quad (20)$$

with w the view angle of the shadowband subtended at the diffuse irradiance sensor, δ is the Sun's declination and I_1 and I_2 are calculated by:

$$I_1 = \cos(\varphi) \cos(\delta) \sin(t_0) + t_0 \sin(\varphi) \sin(\delta) \quad (21)$$

$$I_2 = t_0 \sin^2(\varphi) \sin^2(\delta) + 2 \sin(t_0) \sin(\varphi) \cos(\varphi) \sin(\delta) \cos(\delta) + \cos^2(\varphi) \cos^2(\delta) \left[\frac{t_0}{2} + \frac{\sin(2t_0)}{4} \right] \quad (22)$$

The hour angle at the sunset (t_0) is given from Eq. (10).

4. Results and discussion

The calculated DNI values were validated against measurements by graphical and statistical means under different sky conditions. Based on the sky conditions defined by Eqs. (7a–c), the clear sky conditions cover 80.7% of the dataset, while the cloudy and partly cloudy conditions correspond to 15.0% and 4.3% respectively. The uncorrected and calculated DNI values (by the four models) are presented as a function of the measured DNI for all sky conditions in Figs. 1–4. In the vast majority of cases, the uncorrected DNI is

higher than the measured values. For higher DNI values, corresponding to measurements under cloud-free skies and lower solar zenith angles, there is a smaller dispersion of the points and they tend to be closer to the $y = x$ line. However, for lower measured DNIs, the dispersion of uncorrected DNI values is higher and the overestimation is significant. It is also noticeable that all diffuse corrections models improve the estimation of DNI. The improvement is slight when the Drummond model is applied and the DNI is still overestimated. However, when the LeBaron, the Batlles and the Muneer–Zhang models are used, the improvement is much more

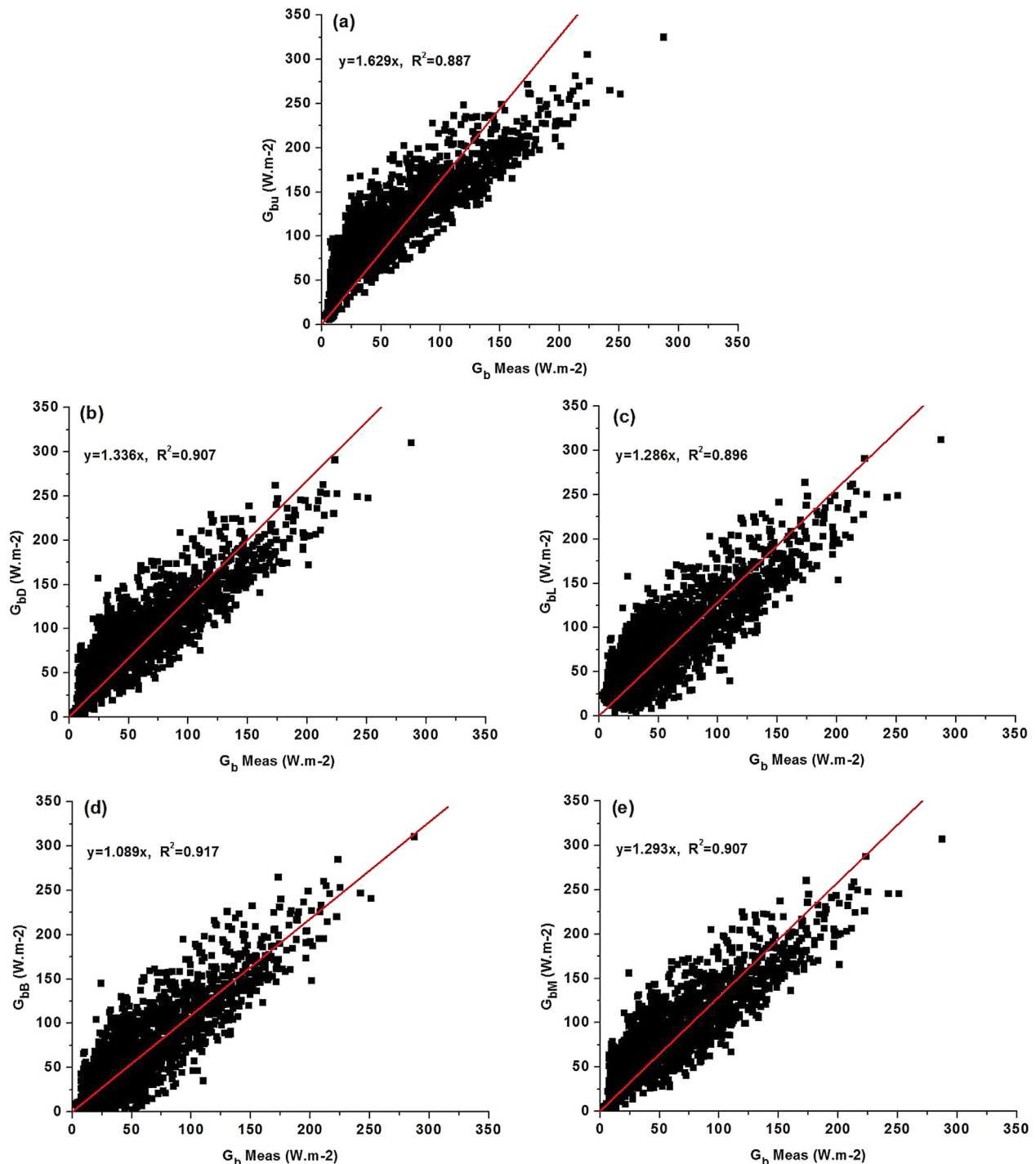


Fig. 4. Same as Fig. 1 but for cloudy conditions.

evident. Scattering in the model-to-measurement comparison graphs remains high for cloudy conditions for all models.

The statistical analysis of the results is presented in Table 2. The average value of the measured DNI and the number of values (or minutes) that correspond to each condition are provided. The slopes corresponding to the LeBaron, the Batlles and the Muneer–Zhang models are closer to unity for all sky and clear sky conditions. For partially cloudy and cloudy conditions all slopes increase significantly with the exception of that of the Batlles model that remains close to unity. LeBaron's slope value is close to unity for all sky and clear sky conditions, decreases for partially cloudy conditions and increases for overcast skies. For the Drummond model, slopes up to 1.029 are obtained for all and clear sky conditions. However, the slope values increase to 1.110 for partially cloudy sky conditions and to 1.336 for cloudy conditions. Despite the high slope values, the Drummond model still improves the results when compared with the uncorrected estimates of DNI. For cloudy conditions, when the DNI values are too low, the best slope (1.089) is obtained using the Batlles model. In this case, the coefficient of determination (R^2) is lower than 0.92 for all models.

The root mean square error (RMSE) of all models is reduced when compared with the uncorrected DNI values. The reduction is higher for all sky and clear sky conditions for the Muneer–Zhang model, where the RMSE is of about 50% lower compared to the respective values of the uncorrected DNI, followed by the Batlles and the LeBaron models. For partially cloudy and cloudy conditions the Batlles model has the smallest RMSE values reducing the dispersion up to almost 60%. High values for rRMSE for cloudy conditions reveal a significant variability and derive from the fact that 1-min averaged data were used. The mean bias error (MBE) values reveal that the Batlles model performs better for all types of sky conditions. The LeBaron model shows the same performance for all cases except for cloudy conditions. Under cloudy skies (when isotropy prevails) the Muneer–Zhang model and the Drummond model perform better than the LeBaron model, where MBE value for the Muneer–Zhang model is only 11% better than that of the Drummond model. MBE values show a slight underestimation of the measured DNI using the LeBaron and the Batlles models and an overestimation using the Drummond and the Muneer–Zhang models. For overcast skies all models overestimate the measured DNI. t -statistic compares the various models and takes into account the dispersion of the results which is not accounted for when the root mean square error and mean bias error are considered separately. The smaller the value of t , the better the model performs. According to the t -statistic results, the Batlles model performs the best in all cases, followed by the LeBaron model for all sky and partially cloudy conditions and by the Muneer–Zhang model for clear sky and cloudy conditions.

An additional analysis is performed by comparing the model residuals as a function of the clearness index k_t (Fig. 5). The residuals (calculated minus measured DNI values) are expressed as a percentage of the average value of the measured DNI. The residuals for the uncorrected DNI are always positive and become maximum (13.8%) under partially cloudy conditions. The residuals of the estimated DNI after applying the Drummond and the Muneer–Zhang models present almost the same behavior; the residuals after applying the Drummond model are about 4% and 0% for cloud-free and cloudy skies respectively but increase for partially cloudy conditions. For the Muneer–Zhang model the residuals reach 0% for clear and overcast skies. When applying the LeBaron model, the DNI residuals are lower than those of the previous models but increase significantly (up to 4%) for $k_t < 0.5$. The lowest residuals ($\pm 1\%$), that are also independent from k_t , are obtained using the Batlles model.

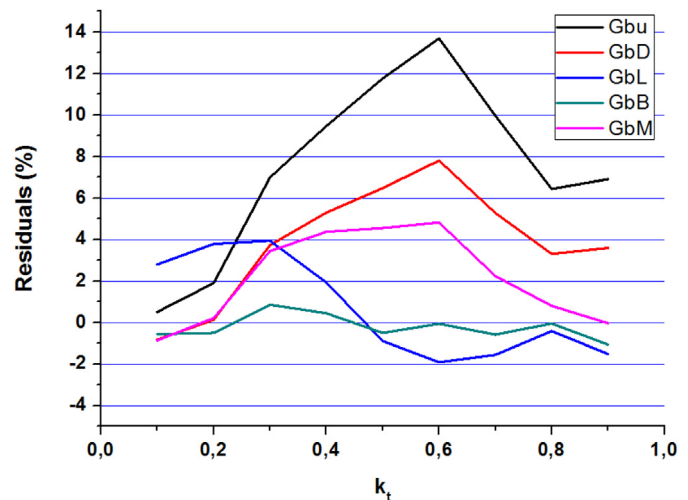


Fig. 5. The residuals of all models (expressed as a percentage of the mean value of G_{bMeas}) as a function of the clearness index k_t .

The distribution of monthly sums of DNI values for the specific site is presented in Fig. 6. As seen from the second (red in the web version) bars, the calculated DNI values (from GHI and DHI measurements) exceed the measured ones (black colored bars). The differences in the calculated monthly irradiance values vary from 9.7 MJ/m² in December to 48 MJ/m² in May. When the diffuse irradiance is corrected with the LeBaron, the Batlles and the Muneer–Zhang models, the amount of calculated direct irradiance approaches the measured one for each month and the differences do not exceed –12 MJ/m², –8.5 MJ/m² and 8.3 MJ/m² respectively. The absolute and percent (%) differences between the calculated and measured values of the annually integrated DNI are presented in Fig. 7. The annual DNI is overestimated by 7.7% (363 MJ/m²) when calculated from the global and diffuse measurements. The calculated DNI using the Drummond and the Muneer–Zhang models is also higher by 4.0% and 1.3% respectively (190.8 MJ/m² and 60.5 MJ/m²) than the measured DNI. However, the calculated DNI using the LeBaron and the Batlles models is slightly lower (–0.8% and –0.4% respectively) than the measured one. The corresponding absolute differences are –36.4 MJ/m² and –19.4 MJ/m².

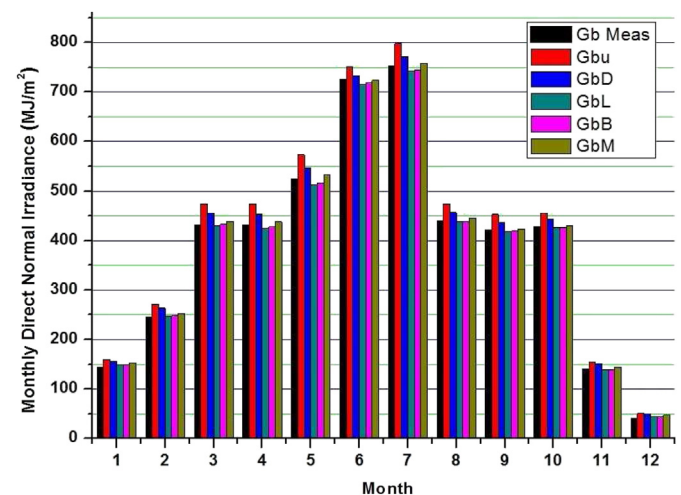


Fig. 6. The distribution of monthly direct normal irradiance using measured, uncorrected and model corrected data.

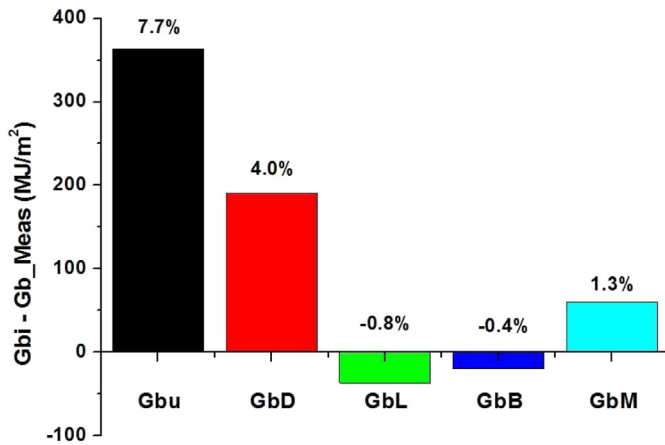


Fig. 7. The absolute and percentage (%) differences between the calculated and measured values of the annually integrated direct normal irradiance.

In many cases [21–26], diffuse correction factors are applied to hourly averaged irradiance data. In order to explore the effect of data averaging on our conclusions, 2178 values of hourly-averaged data were produced for the year 2001 after having been filtered (see Table 1). Then the correction factors were applied to the hourly-averaged data and revealed a similar statistical behavior, shown in Table 3. The Batlles model performs best for all sky types, clear sky and partially cloudy conditions. For cloudy conditions the LeBaron model performs better followed by the Muneer–Zhang model but the sample was very small to draw unambiguous conclusions. Estimation of integrated annual energy differences between the calculated and measured hourly-averaged DNI had the same performance as in Fig. 7 with the percentage differences being 8.1%, 4.1%, –1.1, –0.7% and 1.4% for the uncorrected, Drummond, LeBaron, Batlles and Muneer–Zhang models respectively.

Table 3

Statistical results for the hourly-averaged values of the direct normal irradiance calculations from global and diffuse irradiance data with and without the four shadowband correction models for all, clear, partially cloudy and cloudy sky conditions.

Correction models	Slope	R^2	RMSE (W/m ²)	MBE (W/m ²)	t -Statistic
All sky conditions: $N = 2178$ hourly averages, $\bar{G}_{b, Meas} = 607.93 \text{ W/m}^2$					
Uncorrected	1.059	0.996	57.8 (9.5%) ^a	49.4 (8.1%) ^a	76.77
Drummond	1.030	0.998	35.7 (5.9%)	24.9 (4.1%)	45.59
LeBaron et al.	0.994	0.998	26.9 (4.4%)	–7.0 (–1.2%)	12.72
Batlles et al.	0.997	0.998	25.7 (4.2%)	–4.2 (–0.7%)	7.85
Muneer and Zhang	1.007	0.999	24.9 (4.1%)	8.5 (1.4%)	17.01
Clear sky conditions: $N = 1590$ hourly averages, $\bar{G}_{b, Meas} = 742.57 \text{ W/m}^2$					
Uncorrected	1.052	0.998	52.6 (7.1%) ^a	44.3 (6.0%) ^a	62.24
Drummond	1.026	0.999	33.9 (4.6%)	22.8 (3.1%)	36.33
LeBaron et al.	0.996	0.999	24.4 (3.3%)	–3.9 (–0.5%)	6.52
Batlles et al.	0.998	0.999	24.7 (3.3%)	–1.7 (–0.2%)	2.74
Muneer and Zhang	1.006	0.999	23.4 (3.2%)	5.8 (0.8%)	10.26
Partially cloudy sky conditions: $N = 523$ hourly averages, $\bar{G}_{b, Meas} = 267.15 \text{ W/m}^2$					
Uncorrected	1.178	0.983	73.1 (27.4%) ^a	63.3 (25.2%) ^a	53.66
Drummond	1.088	0.991	42.1 (15.7%)	33.4 (12.5%)	29.75
LeBaron et al.	0.959	0.989	33.9 (12.7%)	–18.1 (–6.8%)	14.40
Batlles et al.	0.979	0.991	29.1 (10.9%)	–10.9 (–4.1%)	9.29
Muneer and Zhang	1.042	0.993	29.7 (11.1%)	17.0 (6.4%)	15.95
Cloudy sky conditions: $N = 65$ hourly averages, $\bar{G}_{b, Meas} = 56.24 \text{ W/m}^2$					
Uncorrected	1.415	0.953	35.9 (63.9%) ^a	31.5 (56.0%) ^a	14.59
Drummond	1.128	0.965	17.3 (30.7%)	9.3 (16.6%)	5.12
LeBaron et al.	1.001	0.931	18.7 (33.2%)	4.1 (7.2%)	1.79
Batlles et al.	0.845	0.918	20.4 (36.3%)	–13.0 (–23.1%)	6.63
Muneer and Zhang	1.08	0.962	15.7 (28.0%)	6.2 (11.0%)	3.41

^a RMSE and MBE as percentage of $\bar{G}_{b, Meas}$.

5. Conclusions

The DHI measured by a pyranometer with an adjusted shadowband must be, a priori, corrected for the obscured portion of the sky. In this study, we use 1-min averaged measurements of GHI, DNI and four widely used models for correcting the DHI measured using a shadowband in Athens, Greece. The accuracy of this correction is important since the subsequent calculation of DNI is affected. The novelty of this study is that, considering the importance of DNI for several solar energy applications, we try to assess the performance of the four diffuse correction models by comparing the calculated DNI with simultaneous pyrhemeter measurements, aiming to the more accurate estimation of DNI at stations where GHI and DHI are measured simultaneously.

The data were divided according to the clearness index k_t value in three sky conditions: clear, partially cloudy and cloudy. The diffuse correction models were applied to the complete dataset and also to the datasets corresponding to the above three sky conditions. It was found that the estimated DNI values using the DHI corrected by the Batlles model are closer to the pyrhemeter measurements. The Drummond's purely geometric factor corrects less the DHI and consequently the derived DNI. The Muneer–Zhang model that corrects Drummond's factor considering the clearness index and the irradiance distribution [4] has similar behavior to that of the Drummond model but has a clearly better performance and overestimates the energy calculated using the DNI values by 1.3% annually. The LeBaron model has similar performance to that of Batlles model underestimating the annual energy calculated by the DNI by 0.8%. When the monthly diffuse irradiance is corrected with the Batlles model, the annual energy calculated by the DNI is slightly underestimated only by 0.4%. The lowest residuals ($\pm 1\%$), with no dependence from k_t , are obtained using the Batlles model.

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